

PROJECT DOCUMENTATION

ComiCraft AI

Comic Story Creator Using Gemini Models

Team Leader:

Madhumitha.R

Team Member:

Meeantchii

Keerthikaa

Kayal

Kaviya Sri

DECLARATION

I hereby declare that this project documentation has been prepared for academic purposes and that the design, explanations, and implementation concepts presented in this report are intended to describe the ComiCraft AI project. Any external technologies, libraries, models, or documentation referenced in the project are acknowledged in the references section.

ABSTRACT

ComiCraft AI is an AI-assisted comic story creator designed to help users transform simple ideas, prompts, characters, themes, and genres into structured comic narratives. The core idea is to combine a user-friendly creative interface with Gemini generative AI models so that users can generate story concepts, character descriptions, scene sequences, dialogues, narration, and visual-storyboard prompts from a single creative input.

Traditional comic creation can require substantial time and skills in writing, character development, scene planning, and dialogue design. ComiCraft AI addresses this challenge by providing a guided workflow in which the user enters an idea and optionally specifies genre, tone, characters, setting, number of scenes, and target audience. The AI processes these inputs and returns a structured comic story that can be edited and refined by the user.

The project follows a modular architecture consisting of a presentation layer, prompt and input-processing layer, Gemini model integration layer, story-generation engine, structured-output processor, and result presentation/export layer. Prompt engineering is used to maintain consistency between scenes and characters while encouraging creative variation.

The proposed system demonstrates how generative AI can support creative writing without removing human control. The user remains responsible for selecting, editing, and refining the generated content. The project can be extended with image generation, speech generation, multilingual support, persistent character memory, collaboration features, and comic-page rendering.

Keywords

Artificial Intelligence, Generative AI, Gemini Models, Comic Generation, Story Creation, Prompt Engineering, Natural Language Processing, Creative AI, Storyboard Generation, Human-AI Collaboration.

TABLE OF CONTENTS

Chapter	Topic	Page
1	Introduction	5
2	Problem Statement and Objectives	6
3	Existing System and Proposed System	7
4	System Requirements and Feasibility	8
5	System Architecture	9
6	Functional Modules	10
7	AI and Gemini Model Integration	11
8	Methodology and Workflow	12
9	Implementation Design	13
10	Testing and Evaluation	14
11	Results, Limitations and Future Scope	15
12	Conclusion and References	15

List of Major Figures

- Figure 1 – High-level ComiCraft AI architecture
- Figure 2 – Comic generation workflow
- Figure 3 – Prompt and response processing flow
- Figure 4 – Proposed user interaction flow

List of Tables

- Table 1 – Functional requirements
- Table 2 – Non-functional requirements
- Table 3 – Module description
- Table 4 – Test cases
- Table 5 – Evaluation criteria

1. INTRODUCTION

1.1 Background

Comics combine storytelling, visual imagination, character development, dialogue, and sequential scenes. Creating a comic traditionally involves several stages: developing a concept, writing the plot, designing characters, planning scenes, writing dialogue, creating artwork, and arranging panels. For beginners, the creative process can become difficult when they have an idea but do not know how to turn that idea into a complete story.

Generative artificial intelligence has introduced new methods for assisting creative tasks. Large language models can understand natural-language instructions and generate coherent text, while multimodal models can support richer interactions involving text and other forms of content. Gemini models can be used as the reasoning and generation layer of applications that need creative text transformation.

1.2 Project Idea

ComiCraft AI focuses specifically on comic-story creation. Instead of returning an unstructured paragraph, the application is designed to produce a comic-friendly structure such as title, premise, characters, scene descriptions, panel actions, dialogue, narration, and ending. The output can therefore serve as a foundation for a comic script or storyboard.

1.3 Need for the System

- Beginners often have ideas but lack a structured storytelling process.
- Writing scenes and dialogues manually can be time-consuming.
- Maintaining character consistency across multiple scenes is difficult.
- A guided AI workflow can accelerate brainstorming and first-draft creation.
- Users need an easy way to regenerate or refine individual parts rather than restarting the entire story.

1.4 Scope

The primary scope is AI-assisted comic story and storyboard text generation. The system can later be extended to visual comic generation, image editing, multilingual output, voice narration, collaborative editing, and publishing.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Many aspiring comic creators struggle to convert a short idea into a complete sequence of scenes. Conventional writing tools provide blank pages but do not automatically guide users through character development, plot progression, scene transitions, or dialogue creation. The project therefore proposes a conversational AI-based solution that converts user ideas into editable, structured comic stories.

2.2 Aim

The aim of ComiCraft AI is to develop an intuitive AI-assisted system that helps users create structured comic stories from natural-language ideas using Gemini models.

2.3 Objectives

1. Accept a simple story idea or prompt from the user.
2. Allow optional configuration of genre, tone, setting, characters, audience, and scene count.
3. Generate a story outline with a clear beginning, development, conflict, and resolution.
4. Generate characters with names, roles, personalities, goals, and relationships.
5. Break the story into scenes and comic panels.
6. Generate natural dialogue and narration appropriate to each scene.
7. Allow users to refine, regenerate, shorten, expand, or rewrite content.
8. Present generated content in a clean, editable format.
9. Maintain reasonable consistency across characters and scenes through structured prompts.
10. Provide an architecture that can be extended to image generation and publishing.

2.4 Target Users

User Group	Possible Use
Students	Creative assignments and storytelling practice
Comic beginners	First-draft scripts and storyboards
Writers	Brainstorming characters and scenes
Educators	Interactive storytelling activities
Content creators	Rapid concept development

3. EXISTING SYSTEM AND PROPOSED SYSTEM

3.1 Existing Approach

In a conventional workflow, a creator develops a concept manually, searches for references, writes character notes, creates a plot, divides it into scenes, writes dialogue, and separately plans the visual composition. Digital drawing and writing applications can improve organization, but they generally require the creator to perform most of the creative planning manually.

3.2 Limitations of Conventional Workflow

- High time requirement for first-draft preparation.
- Difficult for beginners to structure a complete narrative.
- Repeated manual editing when the plot changes.
- Character and scene consistency can be difficult to maintain.
- Separate tools may be required for writing, planning, drawing, and publishing.

3.3 Proposed ComiCraft AI

The proposed application introduces Gemini as an AI generation layer. The user supplies a concept and creative preferences. The system converts those inputs into structured prompts and sends them to the selected Gemini model. The generated response is validated, organized, and displayed as a comic-ready story.

3.4 Comparison

Feature	Conventional Workflow	ComiCraft AI
Idea expansion	Mostly manual	AI-assisted
Character creation	Manual	AI-assisted + editable
Scene planning	Manual	Structured generation
Dialogue	Manual	AI-generated draft
Revision	Manual rewriting	Regenerate/refine prompts
User control	High	High; AI provides suggestions
Visual generation	Separate workflow	Future extension

3.5 Human-in-the-Loop Principle

ComiCraft AI is designed as a creative assistant rather than an autonomous replacement for the creator. Generated content should be reviewed, edited, and approved by the user. This approach helps preserve authorial control while reducing repetitive drafting work.

4. SYSTEM REQUIREMENTS AND FEASIBILITY

4.1 Functional Requirements

ID	Requirement
FR1	User can enter a story idea.
FR2	User can select or enter genre and tone.
FR3	User can define characters and setting.
FR4	System generates a structured comic story.
FR5	System generates scene/panel descriptions.
FR6	System generates dialogue and narration.
FR7	User can refine or regenerate content.
FR8	System displays output in an editable format.
FR9	System can support export in a future version.

4.2 Non-Functional Requirements

Category	Requirement
Usability	Simple interface suitable for beginners.
Performance	Reasonable response time for generation requests.
Reliability	Handle API failures and invalid inputs gracefully.
Security	Protect API credentials and user data.
Maintainability	Use modular components and clear interfaces.
Scalability	Allow additional AI and export features later.

4.3 Hardware Requirements

- Laptop or desktop computer with a modern web browser.
- Minimum 4 GB RAM recommended for development; 8 GB or more is preferable.
- Stable internet connection for cloud model/API access.

4.4 Software Requirements

- Operating system: Windows, Linux, or macOS.
- Modern browser such as Chrome, Edge, Firefox, or equivalent.
- Development environment appropriate to the selected stack.
- Gemini API / Google AI development platform access, subject to the chosen integration method.

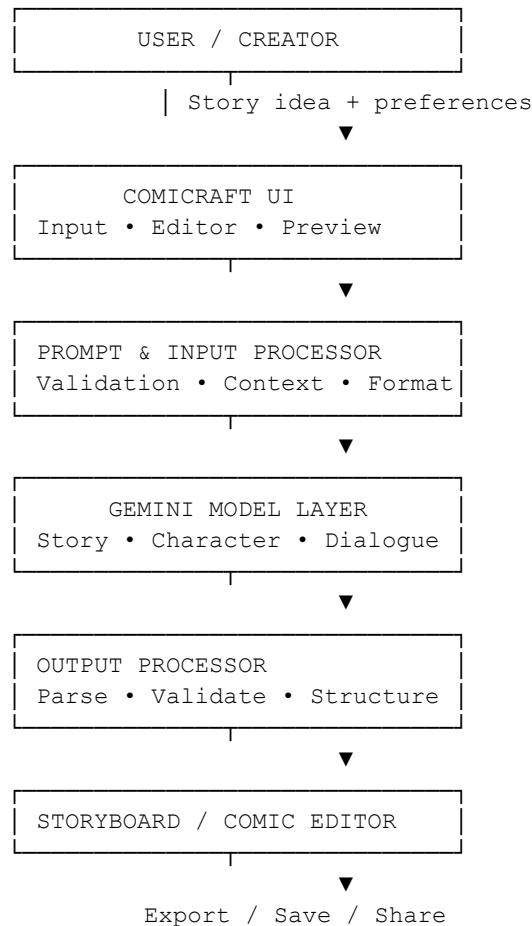
4.5 Feasibility

Technical feasibility is supported by the availability of cloud-based generative AI APIs and standard web-development frameworks. Economic feasibility depends on model usage, API limits, hosting, and storage requirements. Operational feasibility is high for users familiar with basic text-based applications because the system follows a guided input-and-generate workflow.

5. SYSTEM ARCHITECTURE

5.1 High-Level Architecture

The architecture separates the user interface, application logic, AI integration, structured-output processing, and optional storage/export layers. This separation makes the application easier to maintain and extend.



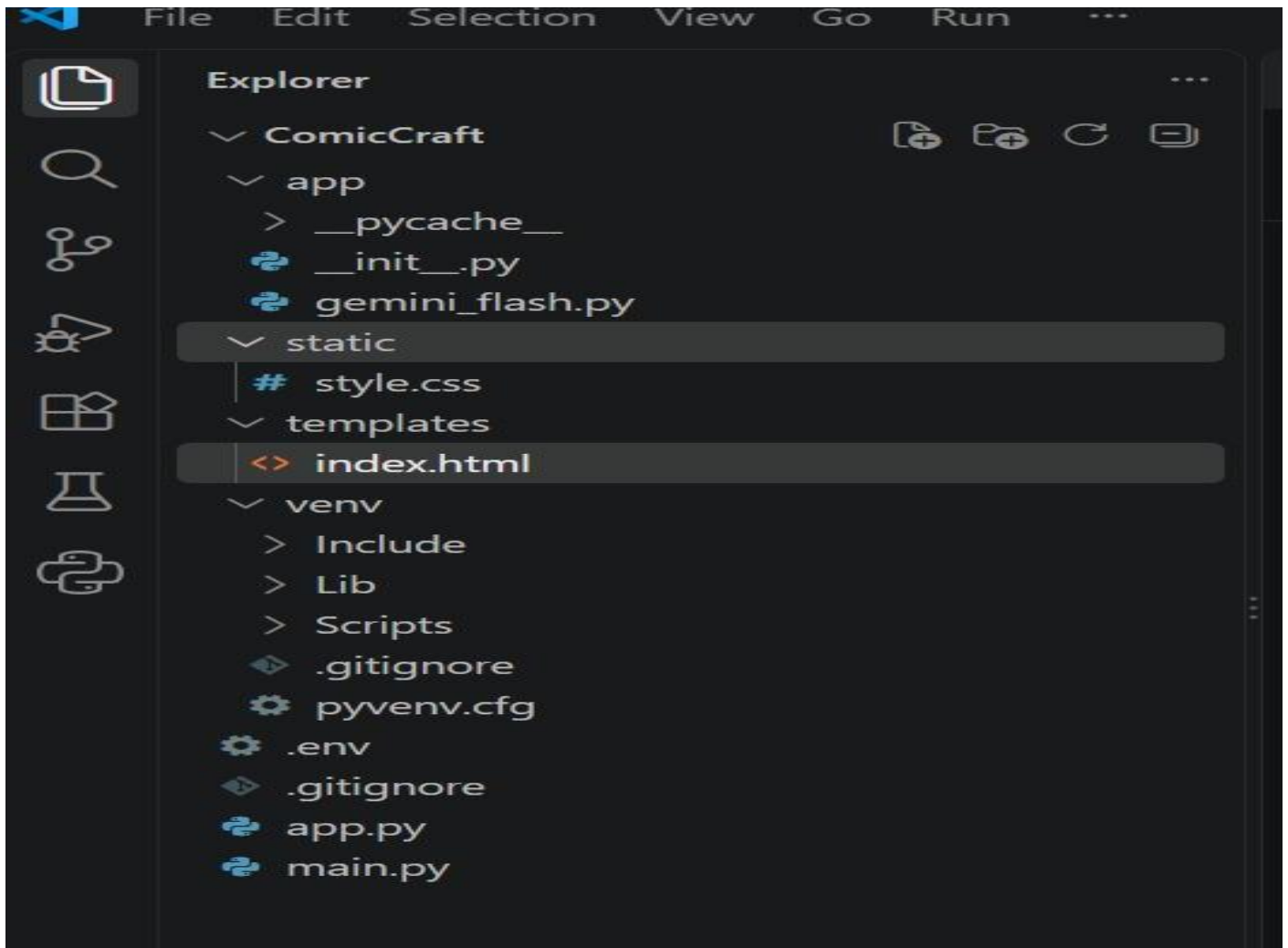
5.2 Main Components

- Presentation Layer – collects inputs and displays generated content.
- Prompt Layer – converts user preferences into clear model instructions.
- Gemini Integration Layer – communicates with the selected Gemini model/API.
- Output Processor – organizes model output into title, characters, scenes, panels, and dialogue.
- Persistence Layer – optional database or local storage for saving projects.
- Export Layer – future support for PDF, text, image, or comic-page formats.

5.3 Data Flow

11. User enters an idea and creative preferences.
12. Application validates required fields.
13. Prompt builder creates a structured instruction.
14. Gemini model receives the prompt and generates content.
15. Application parses and displays the result.

16. User edits, regenerates, or accepts the story.
17. Final content can be stored or exported.



6. FUNCTIONAL MODULES

6.1 User Input Module

This module collects the main story idea and optional settings. Example fields include story title, genre, tone, target audience, setting, main characters, number of scenes, and preferred ending type.

6.2 Story Generator Module

The Story Generator creates the initial narrative structure. It can be instructed to produce a logline, synopsis, plot progression, conflict, climax, and resolution. The result acts as the foundation for subsequent scene generation.

6.3 Character Generator Module

This module develops character profiles. A profile can include name, role, age range when appropriate, personality traits, goals, fears, relationships, strengths, weaknesses, and character arc. The profile can then be reused as context in later generation steps.

6.4 Scene and Panel Generator

The story is divided into scenes and, where required, panels. Each panel may contain visual action, location, characters present, dialogue, narration, mood, and a suggested visual composition. This structured format is useful for later image-generation integration.

6.5 Dialogue Generator

The dialogue module creates character conversations according to the personalities and context defined earlier. The prompt can specify tone, length, language, age suitability, and whether narration should be included.

6.6 Refinement Module

Users can request targeted changes such as “make the ending more emotional,” “shorten scene 3,” “add a plot twist,” “make the dialogue funnier,” or “change the setting.” The system should preserve relevant context while modifying only the requested part.

6.7 Project Management Module

A future project-management layer can allow users to save multiple stories, maintain character libraries, duplicate versions, and restore earlier drafts.

6.8 Export Module

Generated stories can eventually be exported as plain text, PDF, formatted script, or comic-page layouts. Export should preserve headings, panels, dialogue, and narration.

7. AI AND GEMINI MODEL INTEGRATION

7.1 Role of Gemini Models

Gemini models can serve as the generative intelligence behind ComiCraft AI. The application does not need to expose raw model complexity to the user. Instead, it provides a simple creative interface and translates user intent into structured model instructions.

7.2 Prompt Engineering

Prompt engineering is central to the quality of the generated comic. A strong prompt clearly defines the task, context, constraints, desired output format, and creative requirements.

7.3 Example Prompt Structure

SYSTEM ROLE:

You are an assistant that creates structured comic scripts.

INPUT:

Genre: Fantasy

Tone: Adventure with light humor

Setting: Floating city

Main character: A curious student inventor

Scenes: 6

TASK:

Create a coherent comic story with a title, short premise, character profiles, six scenes, panel descriptions, dialogue, narration, and a satisfying ending.

OUTPUT RULES:

Keep character personalities consistent. Label every scene and panel. Keep dialogue concise. Return content in a predictable structure.

7.4 Context Management

For multi-step generation, the application can maintain a compact story context containing the title, premise, character profiles, setting, important events, and previous scene summaries. This context can be supplied to later requests to reduce contradictions.

7.5 Structured Output

Where supported by the selected Gemini integration, structured response formats can be used so that the application receives predictable fields such as title, characters, scenes, panels, dialogue, and narration. Validation should be performed before rendering the content.

7.6 Safety and Responsible Use

- The application should validate and filter inappropriate or unsafe content according to applicable platform policies.
- Users should review generated content before publication.

- The interface should make clear that AI-generated content may contain errors or inconsistencies.
- API keys should never be hard-coded into client-side code or publicly exposed.

8. METHODOLOGY AND WORKFLOW

8.1 Development Method

An iterative development approach is suitable for ComiCraft AI because prompt quality, user interface design, and output structure benefit from repeated testing. The project can be developed in small modules and integrated progressively.

8.2 Workflow

18. Requirement Analysis – identify target users, story inputs, desired outputs, and constraints.
19. UI Design – create a simple interface for story creation and editing.
20. Prompt Design – develop reusable prompt templates for stories, characters, scenes, and dialogue.
21. Gemini Integration – connect the application to the selected Gemini model/API.
22. Output Structuring – parse and validate generated content.
23. Editing Workflow – support targeted refinement and regeneration.
24. Testing – evaluate functionality, consistency, usability, and failure handling.
25. Documentation – record architecture, modules, test cases, limitations, and future work.

8.3 Comic Generation Workflow

Stage	Input	Output
1. Idea	User concept	Story premise
2. Characters	Roles + preferences	Character profiles
3. Plot	Premise + characters	Story outline
4. Scenes	Story outline	Scene sequence
5. Panels	Scene details	Panel descriptions
6. Dialogue	Characters + panels	Dialogue/narration
7. Refinement	User instruction	Updated content
8. Finalize	Approved draft	Comic script/storyboard

8.4 Error Handling

- Handle missing required inputs with clear validation messages.
- Handle API/network errors with retry or graceful fallback messages.
- Validate response structure before displaying generated fields.
- Prevent accidental exposure of API keys and sensitive configuration.
- Allow users to regenerate a failed or unsatisfactory section.

9. IMPLEMENTATION DESIGN

9.1 Suggested Technology Stack

Layer	Possible Technology
Frontend	HTML/CSS/JavaScript or React
Backend	Python / Node.js / equivalent server framework
AI	Gemini model through an appropriate Google AI API/SDK
Storage	Local storage, SQLite, or cloud database
Version Control	Git and GitHub
Export	PDF/text generation libraries

The exact technology stack may be changed according to project requirements, developer experience, hosting environment, and API availability.

9.2 Logical Data Model

Entity	Important Fields
Project	project_id, title, created_at, updated_at
Story	premise, genre, tone, setting, ending
Character	name, role, personality, goal, relationships
Scene	scene_number, location, action, mood
Panel	panel_number, visual, dialogue, narration
Revision	version, change_request, timestamp

9.3 Pseudocode

START

Read user story idea and preferences

Validate required fields

Build structured story prompt

Send prompt to Gemini model

Receive model response

Validate and structure response

Display title, characters, scenes and panels

IF user requests changes:

 build refinement prompt

 regenerate selected section

 update displayed content

Save/export final draft when requested

END

9.4 Interface Design Principles

- Keep the first screen simple and focused on the story idea.
- Use progressive options so advanced controls do not overwhelm beginners.
- Clearly separate generated content from user-edited content.

- Provide regenerate/refine controls near each section.
- Show loading and error states clearly.

10. TESTING AND EVALUATION

10.1 Testing Strategy

Testing should cover both conventional software functionality and AI-specific output quality. A successful API response alone does not guarantee a useful comic. The evaluation therefore includes correctness of the interface, response structure, coherence, consistency, controllability, and usability.

10.2 Functional Test Cases

Test ID	Test Scenario	Expected Result
TC01	Submit valid story idea	Story is generated and displayed
TC02	Submit empty idea	Validation message is shown
TC03	Select genre and tone	Generated story follows requested style
TC04	Generate characters	Character profiles are displayed
TC05	Generate scenes	Scenes follow the story premise
TC06	Regenerate one scene	Only requested content is changed
TC07	API/network failure	Graceful error and retry option
TC08	Very long input	Input is limited/handled appropriately
TC09	Save project	Project is stored successfully
TC10	Export final story	Readable output file is produced

10.3 AI Output Evaluation

Criterion	What to Check
Relevance	Does the output follow the user's idea?
Coherence	Does the story make sense from beginning to end?
Consistency	Do characters remain stable across scenes?
Creativity	Does the output provide useful original variations?
Structure	Are scenes, panels and dialogue clearly organized?
Controllability	Do refinement instructions produce targeted changes?
Readability	Is the result easy to edit and understand?

10.4 Sample Evaluation

A sample evaluation can use several story prompts from different genres such as fantasy, mystery, comedy, science fiction, and slice-of-life. Each generated story can be reviewed against the criteria above. User feedback can then be used to improve prompt templates and interface controls.



ComicCraft

AI Comic Story Creator using Gemini

Create Your Comic

Enter a simple story idea and let Gemini AI create characters, scenes, dialogue and an ending.

A young girl strict a new dress.

🌟 Generate Comic



Your AI Comic

TITLE:

The Beautiful Dress

CHARACTERS:

- A young girl named Maya
- Maya's best friend

SCENE 1:

Description:

Maya gets an idea to create a beautiful dress.

DIALOGUE:

Maya: "I want to design my own beautiful dress!"

Friend: "Let's make your idea come true!"

Description:

Maya chooses colorful fabric and starts designing the dress.

DIALOGUE:

Maya: "This color will make the dress look amazing."

Friend: "Your creativity is wonderful!"

SCENE 3:

Description:

After working carefully, Maya completes the beautiful dress.

DIALOGUE:

Maya: "I created this dress all by myself!"

Friend: "It looks beautiful!"

ENDING:

Maya learns that creativity, patience, and confidence
can turn a simple idea into something special.

YOUR IDEA:

A young girl strict a new dress.

12.1 Conclusion

ComiCraft AI demonstrates a practical application of generative artificial intelligence in creative storytelling. By combining a simple user interface, structured prompt engineering, Gemini model integration, and an editable comic-oriented output format, the system can help users move from an initial idea to a usable comic story draft.

The major strength of the proposed approach is the combination of automation and human control. Instead of treating AI output as a final product, ComiCraft AI positions the model as a creative assistant that can brainstorm, organize, expand, and revise content while the user remains responsible for the final story.

The modular architecture also provides a foundation for future development. Image generation, persistent characters, multilingual support, voice narration, collaborative editing, and automated comic-page layouts can be added without changing the central concept. With appropriate testing, security, content controls, and user feedback, ComiCraft AI can evolve from a story-generation prototype into a broader creative comic-production platform.

12.2 References

- Google AI for Developers – Gemini API and generative AI documentation. <https://ai.google.dev/>
- Google DeepMind – Gemini model family information. <https://deepmind.google/technologies/gemini/>
- Google Cloud – Generative AI and Vertex AI documentation. <https://cloud.google.com/>
- MDN Web Docs – Web development reference for HTML, CSS and JavaScript. <https://developer.mozilla.org/>
- Python Software Foundation – Python documentation. <https://docs.python.org/>
- Git Documentation – Version control documentation. <https://git-scm.com/doc>

12.3 Project Note

This document is a project-report template and design documentation for ComiCraft AI. Specific model names, API methods, interface screenshots, database implementation, performance measurements, and deployment details should be updated to match the actual implementation used by the project team.